POWER PULSE

Measurement of Energy Important Formulae

Notes by Balaji Sir

1 Induction type energy meter

1. $$T_d \propto V\, I \cos\phi, \qquad T_b \propto N \;\Rightarrow\; N \propto P$$

at steady speed driving torque = braking torque, so speed ∝ power

2. $$\text{Revolutions} \;\propto\; \text{Energy} \;\; (\text{meter constant } K \text{ rev/kWh})$$

the number of revolutions is proportional to energy

2 Calibration and error

3. $$E_{true} = P\,t, \qquad E_{rec} = \frac{N}{K}\ \text{kWh}$$

N = revolutions recorded

4. $$\%\text{error} = \frac{E_{rec} - E_{true}}{E_{true}} \times 100$$

positive = meter fast; negative = meter slow

5. $$t = \frac{3600\,N}{K\,P_{kW}} \;\; (\text{seconds for } N \text{ revolutions})$$

used when checking with a stopwatch

3 Features

6. Creeping: rotation at no load (over-compensation); stopped by holes or a slot in the disc